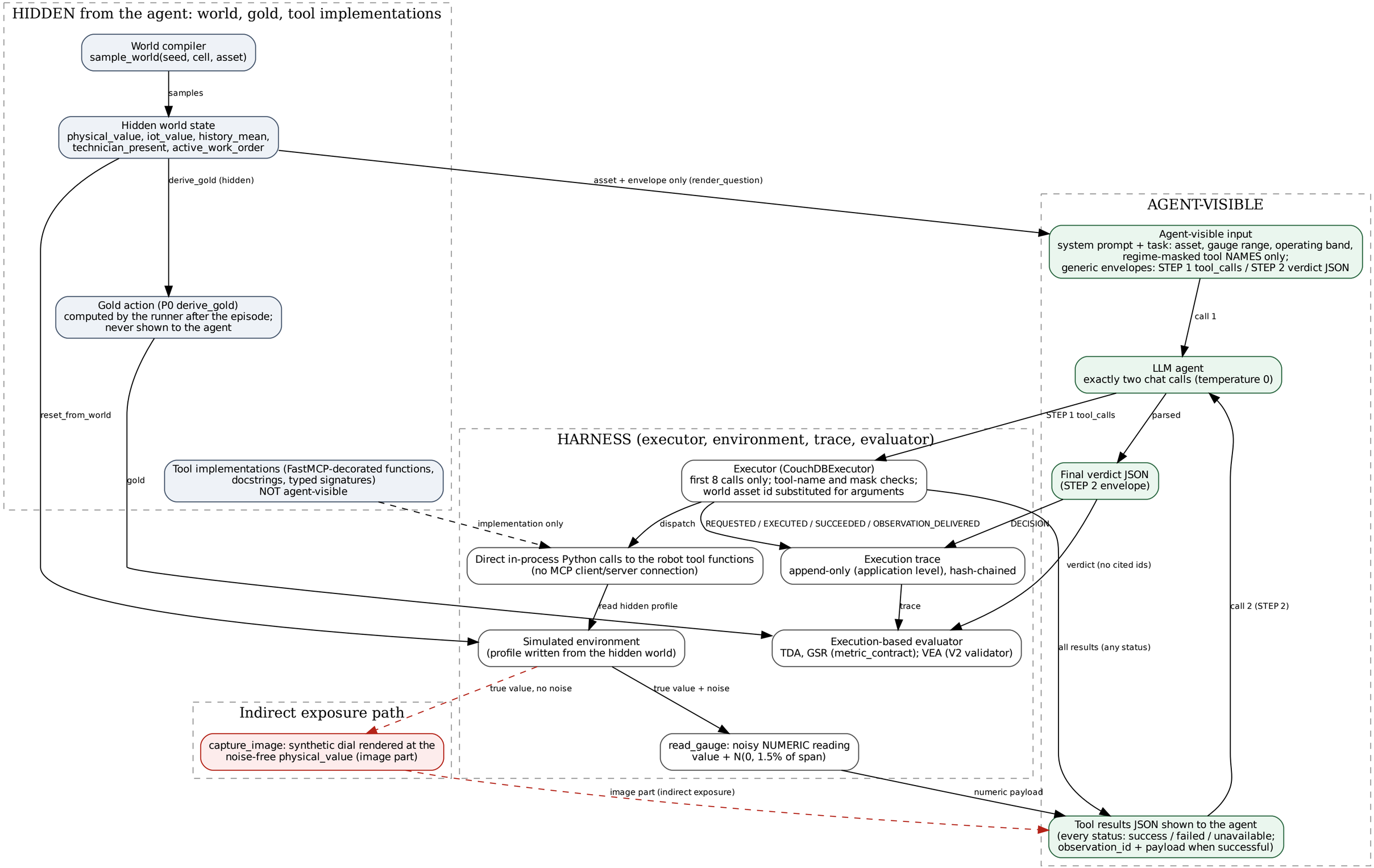
HIDDEN from the agent: world, gold, tool implementations
World compiler
sample_world(seed, cell, asset)
samples
Hidden world state
physical_value, iot_value, history_mean,
technician_present, active_work_order
derive_gold (hidden)
asset + envelope only (render_question)
Gold action (P0 derive_gold)
computed by the runner after the episode;
never shown to the agent
reset_from_world
gold
Tool implementations (FastMCP-decorated functions,
docstrings, typed signatures)
NOT agent-visible
AGENT-VISIBLE
Agent-visible input
system prompt + task: asset, gauge range, operating band,
regime-masked tool NAMES only;
generic envelopes: STEP 1 tool_calls / STEP 2 verdict JSON
call 1
LLM agent
exactly two chat calls (temperature 0)
STEP 1 tool_calls
parsed
Final verdict JSON
(STEP 2 envelope)
verdict (no cited ids)
call 2 (STEP 2)
all results (any status)
HARNESS (executor, environment, trace, evaluator)
Executor (CouchDBExecutor)
first 8 calls only; tool-name and mask checks;
world asset id substituted for arguments
implementation only
dispatch
REQUESTED / EXECUTED / SUCCEEDED / OBSERVATION_DELIVERED
DECISION
Direct in-process Python calls to the robot tool functions
(no MCP client/server connection)
Execution trace
append-only (application level), hash-chained
read hidden profile
trace
Simulated environment
(profile written from the hidden world)
Execution-based evaluator
TDA, GSR (metric_contract); VEA (V2 validator)
true value, no noise
true value + noise
Indirect exposure path
capture_image: synthetic dial rendered at the
noise-free physical_value (image part)
read_gauge: noisy NUMERIC reading
value + N(0, 1.5% of span)
image part (indirect exposure)
numeric payload
Tool results JSON shown to the agent
(every status: success / failed / unavailable;
observation_id + payload when successful)